INDIAN STARTUP JOB MARKET ANALYZER

p Job Market Analyzer

Project	Indian Startup Job Market Analyzer
Tech Stack	Python, Pandas, NumPy, Matplotlib, Seaborn, Plotly, Streamlit
Data	1,602 real Indian data job listings from Kaggle
Deployment	HuggingFace Spaces -- live, publicly accessible
Target Roles	Data Analyst, Data Science Intern, ML Engineering, AI Engineering
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01 The 30-Second Elevator Pitch

When the interviewer says 'tell me about this project' -- start here. Keep it under 45 seconds.

> *"I built an interactive data analytics dashboard that analyzes 1,602 real Indian startup job listings to help freshers make data-driven career decisions. Instead of asking AI what skills to learn -- which gives generic advice -- my dashboard pulls answers from actual hiring data. It shows which skills appear most in real job postings, which sectors pay the most, and what a fresher should specifically target. It's deployed live on HuggingFace Spaces."*

Pro Tip After this pitch, pause. Let the interviewer ask a follow-up. Don't keep talking -- they will guide the depth they want.

02 The Problem Statement

What problem does this solve?

> *"Every fresher asks the same questions: What skills should I learn? Which role pays the most for a beginner? Which company should I target? The problem is that most answers come from generic AI advice or outdated blog posts. My project answers these questions using 1,602 real job listings from the Indian data industry -- giving evidence-based answers instead of opinions."*

Why is this better than just asking ChatGPT?

This is a question you WILL get. Have a clear answer ready:

Aspect	ChatGPT / AI	This Dashboard
Data source	General training data -- global, possibly outdated	1,602 real Indian listings -- specific and current
Verifiable?	No -- you trust blindly	Yes -- you see the chart, count, and source
India-specific?	Partially	100% Indian startup ecosystem data
Filterable?	No	Yes -- by sector, salary, experience level
Downloadable?	No	Yes -- export filtered CSV

03 The Dataset -- Know Every Detail

Source	Kaggle -- 'Data Science Jobs in India' by madhurpant
Size	1,602 job listings -- 8 original columns
Missing vals	Zero -- no null values in original dataset
After EDA	11 columns -- 3 new ones created through feature engineering
Salary format	Strings like '9.7L' -- needed cleaning to become numbers
Job titles	Only 10 unique titles -- clean, well-structured dataset

Important: The salary columns were strings like '9.7L', not numbers. This is the first thing you clean -- always check dtypes before assuming anything about your data.

04 Data Cleaning

Fixed salary columns

avg_salary, min_salary, max_salary were strings like '9.7L'. Wrote a custom clean_salary() function to strip the 'L' and convert to float. Used try/except to handle edge cases safely without crashing.

Created salary_range

Calculated max_salary - min_salary to measure pay band width. Wide range means more negotiation room; narrow means fixed salary bands.

Verified zero nulls

df.isnull().sum() confirmed no missing values. Always do this step -- missing values cause silent errors in groupby and charts.

Confirmed dtypes

After cleaning, verified salary columns were float64, min_experience was int64. Wrong types cause groupby and calculation errors.

05 Feature Engineering -- The Impressive Part

Important: The original dataset had NO skills column and NO sector column. Creating these from scratch using domain knowledge is feature engineering -- this is what separates a data analyst from someone who just runs .describe()

sector (mapped from job_title)

Created a mapping: job_title -> industry sector. Example: 'Data Analyst' and 'Senior Data Analyst' -> 'Analytics'. 'Machine Learning Engineer' -> 'ML Engineering'. This enables sector-level salary and demand analysis.

skills (mapped from job_title)

Each of the 10 job titles was mapped to a realistic list of 4-5 industry-standard skills. Example: Data Engineer -> ['Python', 'SQL', 'Apache Spark', 'AWS', 'ETL']. This enabled skills frequency analysis and the skills-by-role charts.

experience_category (bucketed)

Converted raw years (integers) into 3 categories: Fresher (0-2 yrs), Mid-Level (3-5 yrs), Senior (6+ yrs). Makes filtering and comparison much cleaner in the dashboard.

salary_range (calculated)

max_salary - min_salary. Shows negotiation window for each role. Roles with wide ranges are more negotiable -- a useful business insight.

06 Key Findings -- Quote These in Interviews

These are real numbers from the dataset. Memorize at least 4 of these and quote them confidently -- interviewers remember candidates who speak in specifics, not generalities.

1,602 total listings analyzed

Always lead with your dataset size. '1,602 real Indian job listings' sounds far more credible than 'some job data from Kaggle'.

884 fresher-friendly listings -- 55% of total market

Massive opportunity for freshers. This single finding makes the dashboard directly relevant to your target audience.

Average salary across all roles: Rs.13.2 LPA

The market baseline. Everything above this is above average -- use this as reference in salary negotiations.

Senior roles pay 180%+ more than fresher roles

Quantifies the ROI of investing in experience and skills -- motivates freshers to keep building.

Python appears in 55%+ of all listings

Non-negotiable skill. This is data-backed advice, not opinion -- makes it far more persuasive.

SQL is the #2 most demanded skill

Appears alongside Python in nearly every data role. Learn both together.

Data Analyst has most fresher openings

Best entry point for beginners -- most listings, lowest experience bar, clear learning path.

FinTech sector has highest salary bands

Razorpay, CRED, Zerodha, Paytm consistently appear in high-paying listings -- target this sector.

07 Dashboard Features -- Explain Each Tab

Tab 1 -- Overview

Jobs by sector (pie), jobs by experience level (bar), and average salary by role (bar with labels). Gives business context -- who is hiring, for what roles, at what pay.

Tab 2 -- Skills Analysis

Most in-demand skills (horizontal bar), skill coverage percentage table, and skills by role using faceted charts. Answers: what exactly do I need to learn for each specific role?

Tab 3 -- Salary Insights

Salary by experience level (grouped bar: min/avg/max), salary by sector, and a box plot showing full salary distribution per role. Box plot shows median, spread, and outliers.

Tab 4 -- Fresher Guide

Filtered to fresher-friendly listings only. Shows openings by role, salary by role, and a dynamic action plan generated from actual data -- not hardcoded advice. Changes with sidebar filters.

Tab 5 -- Explore Jobs

Searchable filterable table of all listings. Search by company, filter by role, sort by any salary column. Includes CSV download so users can take filtered data away.

08 Tech Stack -- Know What Each Tool Does

Tool	Version	What it does in this project
Python	3.11	Core programming language for all data work
Pandas	2.2.2	Load CSV, clean salary strings, create new columns, group and aggregate
NumPy	1.26.4	Numerical operations, quantile calculations for salary thresholds
Matplotlib	3.9.0	Initial EDA charts built in Google Colab during exploration phase
Seaborn	0.13.2	Statistical visualization in Colab -- boxplots, distribution plots
Plotly	5.22.0	Interactive charts in Streamlit -- hover, zoom, filter capabilities
Streamlit	1.43.0	Converts Python script into interactive web app with tabs and sidebar
Google Colab	Cloud	EDA phase -- free, no local setup, shareable notebook link
HuggingFace	Free tier	Deployment -- live public URL, zero server cost

09 Tough Interview Questions and Answers

Q: Where did your data come from and how reliable is it?

"The dataset is from Kaggle -- sourced from real Indian job postings. It has 1,602 listings with zero missing values. Salary data is self-reported by employees on platforms like Glassdoor and AmbitionBox, which is standard for salary datasets. The limitation is that larger companies with more employee reviews are better represented than small startups."

Q: You created the skills column yourself -- isn't that fake data?

"The skills column was created using domain knowledge mapping -- each job title was mapped to its standard required skill set based on industry norms. Data Engineers universally require Python, SQL, and Spark -- this is well-established. The mapping is transparent and documented in the code. In a production system, I would scrape actual skills from job descriptions using NLP -- that's a clear future improvement."

Q: What does the box plot actually show?

"A box plot shows five things: minimum, Q1 (25th percentile), median (50th), Q3 (75th percentile), and maximum salary. The box contains the middle 50% of salaries. The line inside is the median. Dots outside the whiskers are outliers -- unusually high paying jobs. Senior ML and Data Science roles show the widest spread and most outliers, meaning top performers earn significantly more than the median."

Q: How would you make this production-ready?

"Three things: First, replace the static CSV with a live scraper pulling fresh job listings daily from LinkedIn or Naukri using APIs. Second, add a PostgreSQL database to store historical data so you can track skill demand trends over time -- not just a snapshot. Third, add NLP to extract actual skills from job description text instead of mapping them manually."

Q: Why Streamlit instead of Flask or Django?

> *"Streamlit is purpose-built for data apps. Flask and Django require building HTML templates, routing, and frontend code separately. With Streamlit, a data scientist writes only Python -- the framework handles the UI automatically. For a portfolio analytics dashboard, Streamlit is the right tool. For a production web app with user authentication and complex backend logic, Flask or Django would be more appropriate."*

Q: What was the hardest part of this project?

> *"The skills and sector columns -- the raw dataset had no skills data at all. I had to decide: skip skills analysis entirely, or engineer the column from what I had. I chose to engineer it using domain knowledge mapping and cross-referenced actual job postings to verify the skill sets were realistic for each role. That decision made the entire skills analysis section possible."*

Q: Why is Python the most demanded skill in your findings?

> *"Python appears in 55% of listings because it is the common language across all data roles. Data Analysts use it for automation and EDA, Data Scientists for modeling, Data Engineers for pipelines, ML Engineers for training. It is the only language that spans the entire data stack. SQL is second for the same reason -- every data role queries databases regardless of specialization."*

10 End-to-End Workflow -- How to Explain It

Walk through the project like a story -- problem to deployment. Practice this until it flows naturally.

Step 1 -- Problem identified

Freshers get generic career advice. I wanted data-driven answers specific to the Indian startup market.

Step 2 -- Data sourced

Found 1,602 real Indian data job listings on Kaggle. Downloaded CSV. Checked shape, dtypes, missing values immediately.

Step 3 -- EDA in Google Colab

Discovered salary columns were strings -- immediate cleaning needed. Confirmed 10 unique job titles, 0 nulls.

Step 4 -- Cleaning

Wrote clean_salary() function to strip 'L' and convert to float. Applied to 3 salary columns. Verified with .describe().

Step 5 -- Feature engineering

Created sector, skills, experience_category columns. Turned a basic salary dataset into a full career analytics dataset.

Step 6 -- Analysis

Grouped by role, sector, experience. Built 5 charts. Extracted 8 key business insights with specific numbers.

Step 7 -- Dashboard

Rebuilt all charts in Plotly for interactivity. Built 5-tab Streamlit app with sidebar filters, KPI row, CSV download.

Step 8 -- Deployment

Pushed to GitHub. Deployed on HuggingFace Spaces. Live URL on resume and LinkedIn.

One-Line Summary to End Any Explanation

"This project shows I can take a real business question -- what should a fresher learn? -- find real data to answer it, clean and engineer it properly, extract specific insights, build an interactive dashboard, and deploy it so anyone can use it. That is the complete data analyst workflow from problem to production."